5 - Creating Agent-Based Models

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Course Outline (Modules)

- Module 1. Introduction to Multi-Agent Systems
- Module 2. Intro to Agent-Based Modeling
- Module 3. On Using ABMs in Cognitive and Computational Social Science
- Module 4. Exploring and Extending Agent-Based Modeling
- **Module 5. Creating Agent-Based Models**
- Module 7. Adaptive Agents
- Module 8. Analyzing Agent-Based Models
- Module 9. More on Analysis of ABMs
- Module 10. Example publication combining ABM, MAS, and Cognition
- Module 11. Group Projects

Creating an agent-based model

- Designing your model
- Building your model
- Examining your model

Designing your model

- What is a model?
  - A conceptual/textual description
  - A software-based implementation

Two major categories of ABMs:

- Phenomena-based modeling
    - Model captures the referent pattern (e.g., segregation, spiral formation, oscillating populations of agents/species)
- Exploratory modeling
    - Create agents, define their behavior, and explore the patterns that emerge

What do research questions look like in the design of models?

- “How does a colony of ants forage for food?”
- No clear question, but want to model X and see what happens

Conceptualizing vs coding models

- *Top-down*
  - Develop entire conceptual model first then implement it
  - Need to have research question, design agents and their rules for behavior, elements of the situation/environment
  - Refine and revise until it has enough detail to be coded

Conceptualizing vs coding models

- *Bottom-up*
  - Choose a domain of interest, start coding something relevant to the domain, adding in the conceptualizations, mechanisms, properties, entities along the way
- In practice:
  - A combination of approaches

ABM design principle

- Start simple and build toward the question you want to answer.
  - Start with the simplest set of agents and behaviors relevant to your topic
  - Always keep your research question in mind (avoid unnecessary stuff)

Verification

- Ensuring that a computational model faithfully implements its target conceptual model
- Simpler models are easier to verify, and scale up from
- Each version of the model should be linked to a research question

Choosing your research questions

- Example for Wolf Sheep Model: “How do the population levels of two species change over time when they coexist in a shared habitat?”
- Make sure the modeling method fits with the questions (agent-based vs. equation based)
  - Heterogeneous agent inhabiting space vs values following a function
- Key aspect is time (how X changes over time)

Exercise: Developing a research question we can answer with an ABM

- Pair up and discuss:
    - You'd like to develop a model to understand how a city council could model transportation patterns of people in the city. What are some important factors here and what kind of research questions could you develop and answer?
    - *Keep track of your key research question*

Choosing your agents

- Agent properties
- Environmental characteristics and stationary agents
- Agent behavior
- Time steps
- Parameters
- Measures

Summary of wolf sheep simple model design

- *Driving Question*: Under what conditions do two species sustain oscillating positive population levels in a limited geographic area when one species is a predator of the other and the second species consumes limited but regenerating resources from the environment?
- *Agent Types*: Sheep, Wolves, Grass
- *Agent Properties*: Energy, Location, Heading (wolf and sheep), Grass-amount (grass)
- *Agent Behaviors*: Move, Die, Reproduce (wolf and - sheep), Eat-sheep (wolf only), Eat-grass (sheep only), Regrow (grass)

Summary of wolf sheep simple model design

- *Parameters*: Number of Sheep, Number of Wolves, Move Cost, Energy Gain From Grass, Energy Gain From Sheep, Grass Regrowth Rate
- *Time Step*:
  1. Sheep and Wolves Move
  2. Sheep and Wolves Die
  3. Sheep and Wolves Eat
  4. Sheep and Wolves Reproduce
  5. Grass Regrows
- *Measures*: Sheep Population versus Time, Wolf Population versus Time

Exercise: Conceptual Model of City Council Transportation

- Using your question from the prior exercise, try to complete the following:
    - *Driving Question*:
    - *Agent Types*:
    - *Agent Properties*:
    - *Agent Behaviors*:
    - *Parameters*:
    - *Time Step*:
    - *Measures*:

Key questions for top-down building your model

1. What part of your phenomenon would you like to build a model of?
2. What are the principal types of agents involved in this phenomenon?
3. In what kind of environment do these agents operate? Are there environmental agents?
4. What properties do these agents have (describe by agent type)?
5. What actions (or behaviors) can these agents take (describe by agent type)?
6. How do these agents interact with this environment or each other?
7. If you had to define the phenomenon as discrete time steps, what events would occur in each time step, and in what order?

Wolf Sheep Model in Netlogo

- Simple 1 has sheep that move around
- Simple 2 sheep require energy to move
- Simple 3 add in regeneration of energy for sheep by eating grass
- Simple 4 add in reproduction to population
- Simple 5 add in wolves

Examining your model

- Multiple runs
  - BehaviorSpace tool in Netlogo, Python code, or R control for netlogo
  - Keep time steps constant but keep random seeds
- Parameter sweeping and results collation
  - Robustness and sensitivity analysis
- Data analysis
  - Statistical inference, visualization, etc.

Exercise: Expanding the Wolf Sheep Model

- The Wolf Sheep Simple model only plots the counts of the animals, but the energy of the animals is almost as important as how many animals there are.
- Right now all of the turtle agents in the Wolf Sheep Simple model move at the same speed. How would you expand the model so that they could move at different speeds?
  - See if you can implement it yourself

Preparing for Module 6 - Components of Agent Based Models

- Complete the reading:
  - CH5, Wilensky & Rand (2015)